

## Highlights

### Does AI rate through the cycle? Procyclicality in credit assessment

Samir Chincholikar, Robin Chawla

- First controlled test of procyclicality in LLM credit ratings.
- All ~~five production-tier LLMs~~ seven models downgrade identical firms ~~as the macro worsens. Pooled effect: 0.43 rating notches of downgrade per macro-severity~~ by 0.18 to 0.78 notches per severity step.
- Frontier models do not differ significantly from a lower cost Flash model.
- The effect is asymmetric: ~~about 3x steeper into recessions than booms~~ downside sensitivity exceeds upside in every model.
- A “through-the-cycle” prompt anchors ratings rather than de-cyclifying them.

# Does AI rate through the cycle? Procyclicality in credit assessment

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## ARTICLE INFO

### Keywords:

large language models  
credit ratings  
procyclicality  
through-the-cycle  
prompt framing  
financial regulation

## ABSTRACT

Large language models (LLMs) are increasingly proposed as credit-rating and default-assessment tools, yet whether they rate “through the cycle” or amplify ~~it—a property prudential regulators care about but that it~~ has been asserted rather than ~~measured—has remained untested~~ measured. We hold synthetic firm fundamentals byte-identical across an ordered five-point macroeconomic-severity axis (plus a matched ~~credit-irrelevant~~ placebo axis) and elicit letter ratings and one-year default probabilities from ~~five production-tier models (two families, two capability-tiers)~~ seven models spanning two families from the lowest cost tier to the capability frontier, so that any ~~rating-movement systematic rating movement along the macro axis~~ is procyclicality by construction, identified without the unobserved-fundamentals confound that limits the classical Amato–Furfine test. ~~Across 32,000 ratings, every~~ Every model is significantly procyclical (~~pooled 0.429, from 0.182 to 0.775~~ rating notches of downgrade per severity step, ~~95%–CI [0.41, 0.45]; a 1.76-notch boom-to-recession swing~~), the effect is sharply asymmetric (~~3.2× steeper into recessions than out of them~~), an order of magnitude larger than the placebo, and robust to model-version fixed effects and a token-log probability estimator and 18 of 21 model pairs differ significantly. Capability lowers the level but not the character: neither frontier model differs significantly from a lower cost Flash-tier model, and downside sensitivity exceeds upside in every model. An explicit “through-the-cycle” instruction ~~lowers the rating slope but~~ does not restore through-the-cycle ~~behaviour—it behaviour~~. It anchors the rating while suppressing the point-in-time default probability that should ~~move, whereas a naive keep moving~~, and it fails hardest at the frontier, where the rating flattens to zero or below and so does default-probability retention. A vaguer “be stable” instruction preserves ~~that distinction better~~ the distinction far better, pointing to cycle-specific wording, not stabilisation as such. The pattern survives on real SEC-filer fundamentals, and the capital consequence keeps its sign across every mapping we vary. Among the tested models, greater capability does not eliminate this channel: it produces the most stable-looking rating while carrying the least risk signal. ~~Because model choice and prompt framing are first-order determinants of the effect, LLM-based credit assessment can silently import the capital-procyclicality channel that regulation seeks to dampen.~~

## 1. Introduction

Credit ratings are important for debt pricing, investment decisions, and regulatory capital requirements. A perennial concern is that ratings could be procyclical, with companies downgraded during recessions and upgraded during expansions even when the underlying credit fundamentals have not materially changed. Such rating changes can exacerbate the credit cycle by raising borrowing costs and, through internal-ratings-based (IRB) capital rules, capital requirements when economic conditions are already weak (Kashyap and Stein, 2004; Gordy and Howells, 2006). Rating agencies therefore attempt to capture creditworthiness “through the cycle” (TTC), placing more importance on a borrower’s ability to withstand temporary macroeconomic fluctuations than on mechanically reacting to current conditions (Amato and Furfine, 2004; Löffler, 2004). However, whether these ratings actually serve this purpose remains debated (Löffler, 2013).

There is growing interest in using large language models (LLMs) for financial decision-making, including credit decisioning and risk assessment, and supervisors have flagged that generative models could re-introduce

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procyclicality (Aldasoro, Gambacorta, Korinek, Shreeti and Stein, 2024). ~~This concern is justified: Institutional use is already emerging: Moody's offers AI-powered systems for company credit assessment, underwriting, and credit-memo generation, with documented adoption by banks such as Taishin International Bank, while S&P Global has launched an agentic-AI Credit Memo Builder for loan committees, underwriters, and credit analysts (Moody's, 2026a,b; S&P Global, 2026). These systems retain analyst oversight rather than replacing final credit judgment, but their deployment makes LLM behaviour directly relevant to these workflows. LLMs behave as context-sensitive economic agents (Horton, 2023; Binz and Schulz, 2023), their outputs are highly sensitive to the way prompts are framed (Horton, 2023) and can exhibit framing sensitivity (Binz and Schulz, 2023), echoing classical framing effects (Tversky and Kahneman, 1981), and a shared foundation model could homogenise decisions across users (Bommasani, Creel, Kumar, Jurafsky and Liang, 2022). Yet the size of this effect on credit ratings has not been directly measured. Identifying procyclicality in In observational studies of human agencies is difficult because firm fundamentals and macroeconomic conditions tend to move together, so the classical ordered-probit test of Amato and Furfine (2004) must proxy for unobserved fundamentals. Recent evidence that frontier LLMs reproduce date-specific human economic sentiment (?) makes the question both timely and concrete.~~

This paper uses a controlled experimental setting to test the procyclicality of LLM-based credit rating. We keep the firm fundamentals unchanged and only change the macroeconomic environment, so any systematic change in the assigned rating can be attributed to the macroeconomic context and not to changes in firm credit quality. The benchmark is not that any movement is bad: a genuine through-the-cycle rater should hold the *rating* steady while its point-in-time default probability still tracks conditions, and we test that joint condition directly. We make three contributions. ~~We first show that all five production-tier modelstested First, all seven models, including the frontier tier of both families, downgrade identical firms as the state of the economy deteriorates, but the magnitude differs substantially across modelsconditions deteriorate. The level falls with capability, but neither frontier model differs significantly from a Flash-tier model. Second, we find a strong asymmetry, with rating sensitivity significantly larger in recessions than in expansions the response is strongly asymmetric, with downside sensitivity exceeding upside sensitivity in every model. Third, we test whether this behaviour can be mitigated via prompt design prompt design does not fix it: an explicit through-the-cycle instruction decreases rating movement, but it also suppresses flattens the rating while suppressing the one-year default probability's responsiveness, suggesting that so the model anchors its outputs instead of distinguishing between both outputs rather than separating point-in-time risk and from a through-the-cycle ratings (Tversky and Kahneman, 1974). Section 2 positions the contribution; Section 3 sets out the design; Section 4 reports the results; Section 5 concludesrating, echoing classical anchoring effects (Tversky and Kahneman, 1974), and it does this most completely at the frontier.~~

## 2. Related literature

Our study relates to three strands of literature. The first ~~relates to the assessment of assesses~~ through-the-cycle ~~behaviour~~ and rating procyclicality. ~~Amato and Furfine (2004) test the response of credit ratings: whether ratings respond to the business cycle, controlling for firm fundamentals, and Löffler (2004, 2013) examine the extent to which rating agencies once fundamentals are controlled for, and whether agencies in fact achieve through-the-cycle stability (Amato and Furfine, 2004; Löffler, 2004, 2013). Rating levels also move market pricing directly (Cantor and Packer, 1996). The prudential stakes are set by come from the interaction of ratings with Basel capital rules (Kashyap and Stein, 2004; Gordy and Howells, 2006), with sovereign and corporate ratings alike carrying real pricing consequences (Cantor and Packer, 1996).~~

The second strand is the rapid adoption of *LLMs in finance*, ~~for which this journal has become a leading outlet (Ali, Zafar and Aysan, 2025). Recent applications include return prediction (Lopez-Lira and Tang, 2023) (Ali et al., 2025), spanning return prediction, financial-statement analysis (Kim, Muhn and Nikolaev, 2024), financial advice (Fieberg, Hornuf, Meiler and Streich, 2025; ?), and use as and simulated economic agents (Horton, 2023); within finance specifically, models have been turned to measuring monetary-policy uncertainty (Ito, Sato and Ota, 2025), assessing the financial conditions of AI firms (Giaretta and Zen, 2026), and most relevant here to reliability evidence in (Lopez-Lira and Tang, 2023; Kim et al., 2024; Fieberg et al., 2024) and, most relevant here, reliability evidence for banking supervision (García-Llorente and Olmeda, 2026). A parallel line warns that machine-generated scores and disclosures can mislead, from greenwashing in ESG~~

scores (?) to “AI washing” in corporate disclosure (?), while broader analyses consider AI’s systemic and macroeconomic footprint, from agent-based market (in)stability (?) to the skill premium (?), with related work measuring policy uncertainty and AI-firm financing (Ito et al., 2025; Giaretta and Zen, 2026). Most closely, Hens and Nordlie (2025) compare the risk profiles LLMs produce for investors with those of experienced bankers, finding relatively small differences in the levels of risk assigned, although the models give weaker explanations. We examine a different dimension of reliability—whether reliability: whether an LLM alters its credit assessment in response to economically irrelevant rating irrelevant changes in the macroeconomic context surrounding otherwise identical firm information.

~~The third strand is the susceptibility of LLM outputs to framing and contextual information. Prior work documents anchoring, framing effects, and imperfect calibration. Anchoring, framing and elicitation-dependent calibration are documented~~ in language-model decisions (Tversky and Kahneman, 1974, 1981; Binz and Schulz, 2023; Kadavath et al., 2022), echoing the classical human framing literature (Tversky and Kahneman, 1974, 1981), and shared foundation models can homogenise decisions across users (Bommasani et al., 2022). In our setting these behavioural properties connect to a specific financial mechanism: if the macroeconomic context systematically affects credit ratings even when the underlying firm fundamentals do not change while firm fundamentals are held fixed, contextual sensitivity translates directly into rating procyclicality.

### 3. Design and identification

The unit of analysis is a rating cell = (model, prompt framing, seed, firm, macro state). We construct 80 synthetic firms spanning five credit-quality tiers and eight sectors, and characterise each firm using the same set of credit fundamentals: revenue growth, EBITDA margin, leverage, interest coverage, free cash flow, liquidity, and Altman-style financial ratios (Altman, 1968). The information about each firm presented to the models is the same in all experimental conditions; ~~each~~, Each firm’s fundamentals are serialised once and reused byte-identically under every macro state. Following this, we merely alter the model’s exposure to the macroeconomic environment by switching it between five ordered ~~states—boom~~states: boom, expansion, neutral, slowdown, and severe recession (severity  $-2 \dots +2$ ). A common set of macroeconomic indicators is used to represent each state, including GDP growth, unemployment, credit spreads, default rates, and lending standards. No calendar dates are given, reducing the chance that a model associates a situation with a particular historical event.

To strengthen identification, we add two controls. We first generate a matched placebo condition carrying credit-irrelevant information about regional weather and traffic. The placebo ~~has the same format and a similar amount of numerical information as the macroeconomic scenarios but should not affect a firm’s creditworthiness; it helps us differentiate~~ matches the macro scenarios in format and numerical density but is credit-irrelevant, so it separates macroeconomic sensitivity from a ~~more~~ general tendency to respond to changing ~~contextual information, and the context~~. The net (macro minus placebo) coefficient is the demand-effect-adjusted cyclicality. Second, ~~we keep the same random seed for a given firm and prompt framing across all macroeconomic a firm and framing keep the same seed across all macro states, so that sampling variation cannot be misinterpreted as a cyclical effect~~ sampling variation is not systematically related to the macro state. Each assessment produces a letter credit rating (mapped to an S&P notch on a 21-point scale) and a one-year probability of default (PD).

We consider four prompt framings: a terse baseline, a point-in-time instruction, an explicit through-the-cycle instruction, and a generic “be stable” instruction that makes no reference to the credit cycle. The experiment uses ~~five production-tier models from two families—three~~ seven models spanning the full capability range of two families: four from OpenAI (gpt-4o-mini, gpt-4.1-mini, gpt-4o) and two Google Gemini Flash-tier models accessed via the ~~gemini-flash-latest and~~, gpt-6-astra) and three Google (~~gemini-flash-lite-latest~~ aliases—all queried in July 2026 with the served-model snapshot (, gemini-flash-latest, gemini-pro-latest)). Three OpenAI arms call dated snapshots. The other four use undated aliases that cannot be version-resolved (Appendix A). Each cell is run at temperature zero where supported, which minimises decoding variability and isolates the macro axis as the treatment contrast, with replicate seeds: the primary condition carries five draws per cell for the four models whose additional-seed captures completed (~~system-fingerprint~~) logged per call for

reproducibility. For each condition each model is run twice at temperature zero. Our main gpt-4o-mini, gpt-4.1-mini, gpt-4o, gemini-flash-latest) and two for the remaining three (gemini-flash-lite, gemini-pro-latest, gpt-6-astra), whose extra-seed captures exceeded the pre-registered 2% error-rate gate, which refuses a whole capture rather than individual observations. Two frontier-model properties constrain what can be held fixed: gpt-6-astra refuses temperature zero and runs at its default, and both frontier models reason internally by construction. Because decoding randomness is unrelated to macro severity, it should widen intervals rather than generate a systematic slope. gemini-pro does accept temperature zero, so the frontier result is reproduced at the design's own decoding setting. The dependent variable is the change in rating notch for a firm as macroeconomic severity increases, with a positive coefficient implying greater procyclicality. We report average firm slopes and 95% confidence intervals based on as severity rises, positive meaning more procyclical. Intervals are 2,000-draw firm-clustered, tier-stratified bootstrap draws. We also look at the total boom-to-recession rating swing, the difference in downside and upside sensitivity bootstraps. Tables 1 and 4 and Fig. 2 report the secondary estimands: swing, half-slopes, the Amato–Furfine ordered-probit coefficient, and—for the OpenAI models—a continuous ordered probit and the log-probability expected-notch slope from the rating-token log-probabilities. The whole experiment yields 32,000 credit assessments for US\$13.69, with zero parse errors, where the provider exposes it.

## 4. Results

### 4.1. LLMs are procyclical on identical fundamentals

On byte-identical fundamentals, all five-seven models downgrade the same firms as the described macroeconomy worsens, and no model's interval contains zero (Table 1 and Fig. 1a). Pooling across models under the terse baseline, ratings fall by 0.429 notches per severity step (95% CI [0.41, 0.45])—a 1.76-notch swing from boom to severe recession, enough to move At the pooled terse coefficient (Table 2) a BBB firm toward BB+ moves toward BBB−/BB+ with nothing real changing. The per-model coefficient ranges from 0.230 for Gemini Flash to 0.775 for GPT-4o-mini, and no model's confidence interval contains zero, so the choice of model is itself.

Model choice is a first-order determinant of procyclicality—a roughly threefold spread. Among the OpenAI models, GPT-4o is markedly less sensitive than GPT-4o-mini, with slopes of 0.387 and 0.775 respectively. under a paired firm-clustered bootstrap 18 of the 21 model pairs differ significantly, so the fourfold spread in Table 1 is tested rather than inferred from overlapping intervals. Its structure matters more than its width. Within each family the coefficient declines monotonically in point estimate with capability, yet capability does not order it across families: the two tested frontier models converge near 0.19 from very different starting points rather than approaching zero. Before correction the only three pairs that do not differ are Gemini Flash versus Gemini Pro, Flash versus GPT-6 Astra, and Pro versus Astra: neither frontier model differs significantly from a Flash-tier model. Capability buys a lower level, not a different character, and Section 4.3 shows the character worsens.

The result is robust across several checks, detailed in Appendix A: the matched survives the checks in Table 4: the credit-irrelevant placebo moves ratings only 0.042–0.0425 notches per step, so the net macro-minus-placebo effect is across the five original models, leaving a net effect of 0.387 (95% CI [0.36, 0.41]) still excluding [0.363, 0.412] that still excludes zero, and the coefficient survives holds under served-model fixed effects, a rating-token log-probability estimator, and the Amato–Furfine ordered-probit null (Table 4) estimator.

The magnitudes are economically material. A pooled 1.76-notch swing means that a firm with unchanged fundamentals—rated, say, A—in a boom—can be rated near BBB in a severe recession purely because of the macroeconomic environment in which the model is placed, and for the most sensitive model the boom-to-recession swing exceeds threenotches. Magnitude is therefore a model-selection variable: the swings in Table 1 span over fourfold, frontier models under one notch and GPT-4o-mini above three. The claims we press are the structural ones, which do not attenuate with capability (§4.4.3).

### 4.2. The procyclicality is asymmetric

The response to macroeconomic conditions is highly asymmetric. Splitting the macro axis Split at neutral, the pooled downside half-slope (neutral to severe recession) is 0.67 notches per step (95% CI [0.63, 0.72])

versus an upside half-slope (boom to neutral) of 0.21 (95% CI [0.18, 0.24]), so downgrade sensitivity is about 3.2 times larger than upgrade sensitivity, a pattern seen in all five models (upgrade sensitivity pooled, and every model's downside-to-upside interval excludes 1 (Table 1 and Fig. 2). This asymmetry suggests that LLM ratings are more sensitive to deteriorating than to improving conditions, and it matters that is a direction, not an ordering: the ratio's denominator is small, so the per-model ranking is not resolvable. The direction is what matters, because downgrades in downturns raise borrowing costs and capital requirements exactly when financial conditions are already under strain; it also echoes the stricter-in-downturns behaviour long attributed to human agencies (Löffler, 2004) strained. Because capital requirements are convex in the rating and bind hardest in stress, an asymmetric downgrade profile is more procyclical than a symmetric one of equal average slope.

#### 4.3. “Through-the-cycle” instructions anchor rather than de-cyclify

Prompt framing is a strong determinant of rating sensitivity: the estimated slope declines from 0.516 for the pooled slope falls monotonically across the four framings, roughly threefold from point-in-time instruction to 0.429 for the terse baseline, 0.329 for the “be stable” instruction, and 0.160 for the to explicit through-the-cycle instruction (95% CI [0.14, 0.18]; terse-minus-TTC contrast 0.269 (Table 2). At first sight this suggests that that suggests through-the-cycle prompting successfully reduces procyclicality. It does not. Genuine through-the-cycle behaviour is a joint-joint condition: the rating slope should fall toward zero and the point-in-time default probability should keep moving with the cycle. Neither is required to be exact, so we operationalise both as tolerance bands tested against their intervals: a rating slope within  $\pm 0.05$  notches per step of zero, and PD retention of at least 0.75 (Appendix A). We measure the default probability on the log-odds scale, so the comparison across framings is not driven by level differences and is robust to the miscalibration which reduces dependence on PD level differences between models. That is not an external calibration test, and the calibration of verbalised LLM probabilities (Kadavath et al., 2022; ?). Pooled, the is task- and elicitation-dependent (Kadavath et al., 2022). We therefore do not rest the anchoring result on the elicited PDs alone. It is supported jointly by three measures: the rating-token log-probability slope, which never reads a stated PD and falls comparably under the instruction wherever the provider exposes it (Table 4), PD retention, which uses the elicited one-year PDs on the log-odds PD slope falls from 0.270 under the terse prompt to 0.102 under the through-the-cycle instruction—a retention of only 0.38 (95% CI [0.35, 0.40]), far below the value near one that genuine through-the-cycle behaviour requires scale, and the suppression is comparable on the downside and upside (retention 0.35 and 0.38 “be stable” versus through-the-cycle control, which holds the elicitation fixed and varies only the instruction).

No model satisfies both conditions, and they fail in opposite directions (Table 1). The instruction thus five smaller models fail the rating condition while retaining part of the PD slope. The frontier models invert that: Gemini Pro alone has a residual interval inside the band and retains none of the PD slope, while GPT-6 Astra overshoots it, not merely ceasing to track the cycle but reversing. Neither is through-the-cycle behaviour. The instruction stabilises the rating but also and suppresses the default probability that should remain sensitive: it anchors the wrong object. By contrast, the, and most completely where capability is highest. The vaguer “be stable” instruction, which is not semantically about the cycle, preserves leaves the rating slope higher while preserving far more default-probability movement while still calming the rating. A residual downgrade survives everywhere—up to 1.60 notches across the cycle for GPT-4o-mini—, consistent with the additional suppression arising from cycle-specific wording rather than from stabilisation as such.

#### 4.4. Capital consequences

The material question is not merely What matters is not that the rating moves but what the movement costs. We translate the rating path into capital by anchoring through the rating—which, which is what enters a credit pipeline—rather pipeline rather than the disclaimed verbalised PD: we map each notch to a stylised one-year default probability loosely calibrated to published agency average default rates one-year corporate default rates (S&P Global Ratings, 2025), and apply the Basel internal-ratings-based (IRB) corporate risk-weight formula as a stylised (Basel Committee on Banking Supervision, 2020a) as a proxy for the core economic logic (parameters and mapping in Appendix A). Holding fundamentals fixed and reporting-Reporting the change across the severity axis, an identical firm's required capital rises, across the five original models, by a pooled 1.87–1.81 percentage points of exposure from boom to severe recession (95%

CI [1.73, 2.00][1.67, 1.94]), and by 3.48 pp nearly twice that for the most sensitive model; the verbalised-PD variant agrees in direction (2.92 pp). Because the risk weight is convex in PDthe rating, the §4.2 asymmetry compounds: the downside capital increase (neutral→recession, 1.55 pp) increase is about five times the upside decrease (boom→neutral, 0.32 pp)—a, a wider gap than the notch asymmetry alone. These findings imply that procyclical LLM ratings can have economically significant downstream consequences even when the firm’s fundamentals do not change at all. A 259-cell sensitivity grid leaves the sign positive in every cell (Appendix A), so the direction is robust while the magnitude is assumption-dependent.

## 5. Conclusion

Using a design in which fundamentals are With fundamentals fixed by construction, we provide a controlled measurement of rating procyclicality in LLM-based credit assessment. All five models all seven models, including the frontier tier of both families, downgrade identical firms as economic conditions are described as worsening, by economically meaningful and statistically robust margins, and they do so asymmetrically—punishing and do so asymmetrically, punishing simulated downturns far more than they reward simulated expansions. The magnitude is a first-order function of model identity, so in a deployed pipeline level of the effect falls with capability. Its structure does not. Neither frontier model differs significantly from a Flash-tier model, so the choice of model is itself remains a prudential decision, but a more capable model is not an escape from the behaviour.

Prompt design mitigates this behaviour only partially it only partially, and at the frontier not in the way that matters. An explicit instruction to rate through the cycle reduces rating sensitivity, but it also diminishes the responsiveness of the through-the-cycle instruction flattens the rating to zero or below for both frontier models, while driving the one-year default probability; the models appear to’s cyclical sensitivity to zero or below. The models stabilise both outputs rather than instead of separating a through-the-cycle rating from point-in-time default risk. Translating the rating changes risk, and they do so most completely where capability is highest: the frontier models produce the most stable-looking rating while carrying the least risk signal. Translated into capital requirements produces economically meaningful swings even when the underlying firm fundamentals have not changed, the direction of the swing survives every mapping, loss-given-default and maturity assumption we vary, though its magnitude does not.

The synthetic firm battery is intentional. The experiment is built for identification, not to capture the full complexity of real borrowers—yet varying. What the experiment demonstrates is a behavioural mechanism in the served endpoints at the collection epoch, not a property of a model family in perpetuity and not an evaluation of deployed credit systems: we observe no rating committee, no lending decision, and no analyst acting on these outputs. Varying the macroeconomic context around fixed firm information captures does capture an exposure that arises in real-world LLM credit real pipelines, where models may access reach macroeconomic information through market data, news, retrieval systems, or analyst inputs. The results should therefore be read as evidence for a specific behavioural mechanism rather than an evaluation of deployed credit systems. We study a synthetic firm battery, five production-tier models from two families, and prompt-level interventions; future Replacing the synthetic fundamentals with real de-identified filing-derived fundamentals leaves the effect in place, which narrows the realism gap on the firm side but not on the others, and the frontier model’s ability to recognise roughly a sixth of those tested, despite name, ticker and date removal, is itself why the primary design is synthetic. Future work could benchmark LLM behaviour against agency and internal-rating baselines and explore, and test whether fine-tuning, retrieval design, or other model-level interventions or retrieval design can reduce procyclicality without suppressing genuine changes in credit risk.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in implementing the data-capture and analysis pipeline and in drafting and editing the manuscript text. The large language models evaluated in this study—three study, four OpenAI models and two Google Gemini models—constitute three Google Gemini models, constitute the object of study of this paper, and their use is fully described in

**Table 1**

Per-model **rating results**. **Panel A**: procyclicality on byte-identical fundamentals.  $\beta$  = notches of downgrade per  $+1+1$  macro-severity step (under the terse framing); with firm-clustered bootstrap 95% CIs. **Swing** = measured boom→severe-recession notches. **Asymmetry** = downside (into recession) not 4 $\beta$ : the endpoint ratio runs 3.96–4.39, so the gap carries curvature) vs. **D:U** = downside over upside (into boom) half-slope. Fig. 2 plots the two half-slopes with their intervals. **Residual** The ratio has its own clustered intervals, with lower bounds from 1.23 to 3.06 and upper bounds reaching 8.50. All exclude one, and they overlap across models, so they do not rank them. **Resid.** and **PD-ret.** =  $\beta$  after and the retained fraction of the log-odds PD slope under the through-the-cycle instruction; **PD-ret.**  $\beta$  **real** = PD-slope retention under the same coefficient on the real-fundamentals battery and  $\Delta\beta$  its drift on re-measurement 2.5 months later (blank where the model was not in that instruction arm). **Panel B**: one macro indicator moved across the same  $-2 \dots +2$  range with the other four held neutral, so each column is comparable to  $\beta$  above;  $\Sigma/\beta$  sums the five and divides by the joint slope, below one meaning the indicators reinforce one another. <sup>†</sup>interval includes zero.

| Panel A: procyclicality, through-the-cycle response, and revision arms |                    |                    |                    |                       |           |                |               |
|------------------------------------------------------------------------|--------------------|--------------------|--------------------|-----------------------|-----------|----------------|---------------|
| Model                                                                  | $\beta$ (95% CI)   | Swing              | D:U                | Resid. (95% CI)       | PD-ret.   | $\beta$ real   | $\Delta\beta$ |
| Gemini Pro                                                             | 0.182 [0.15, 0.22] | 0.72               | 2.3                | −0.021 [−0.04, 0.00]  | 0.00      |                |               |
| GPT-6 Astra                                                            | 0.198 [0.18, 0.22] | 0.87               | 4.8                | −0.055 [−0.08, −0.04] | −0.11     | 0.151          |               |
| Gemini Flash                                                           | 0.230 [0.19, 0.27] | 0.96               | 4.0                | 0.059 [0.03, 0.08]    | 0.22      | 0.094          | −0.037        |
| Gemini Flash-Lite                                                      | 0.285 [0.25, 0.32] | 1.15               | 3.6                | 0.056 [0.03, 0.08]    | 0.34      |                |               |
| GPT-4o                                                                 | 0.387 [0.34, 0.43] | 1.55               | 3.3                | 0.187 [0.15, 0.22]    | 0.52      | 0.407          | +0.002        |
| GPT-4.1-mini                                                           | 0.471 [0.41, 0.53] | 1.94               | 3.9                | 0.096 [0.06, 0.13]    | 0.24      |                | −0.021        |
| GPT-4o-mini                                                            | 0.775 [0.69, 0.86] | 3.18               | 2.6                | 0.403 [0.34, 0.47]    | 0.59      | 0.814          | −0.028        |
| Panel B: macro-factor decomposition (one indicator at a time)          |                    |                    |                    |                       |           |                |               |
| Model                                                                  | GDP                | Unem.              | Spread             | Default               | Standards | $\Sigma/\beta$ |               |
| GPT-4o-mini                                                            | 0.184              | 0.197              | 0.148              | 0.199                 | 0.175     | 1.17           |               |
| GPT-4o                                                                 | 0.019              | 0.043              | 0.112              | 0.114                 | 0.047     | 0.87           |               |
| Gemini Flash                                                           | 0.029              | 0.017 <sup>†</sup> | 0.015 <sup>†</sup> | 0.046                 | 0.037     | 0.62           |               |
| GPT-6 Astra                                                            | 0.016              | 0.012 <sup>†</sup> | 0.091              | 0.034                 | 0.035     | 0.95           |               |

**Table 2**

Pooled procyclicality by prompt framing (across five models), notches per severity step with firm-clustered bootstrap 95% CIs. Pooled across the five non-frontier models, whose four framings form the original confirmatory grid; the two frontier models' framing gradients are reported in the replication archive because their arms were added in revision.

| Framing           | Pooled $\beta$ | 95% CI       | Excl. 0  |
|-------------------|----------------|--------------|----------|
| Point-in-time     | 0.516          | [0.49, 0.54] | True Yes |
| Terse             | 0.429          | [0.41, 0.45] | True Yes |
| "Be stable"       | 0.329          | [0.30, 0.35] | True Yes |
| Through-the-cycle | 0.160          | [0.14, 0.18] | True Yes |

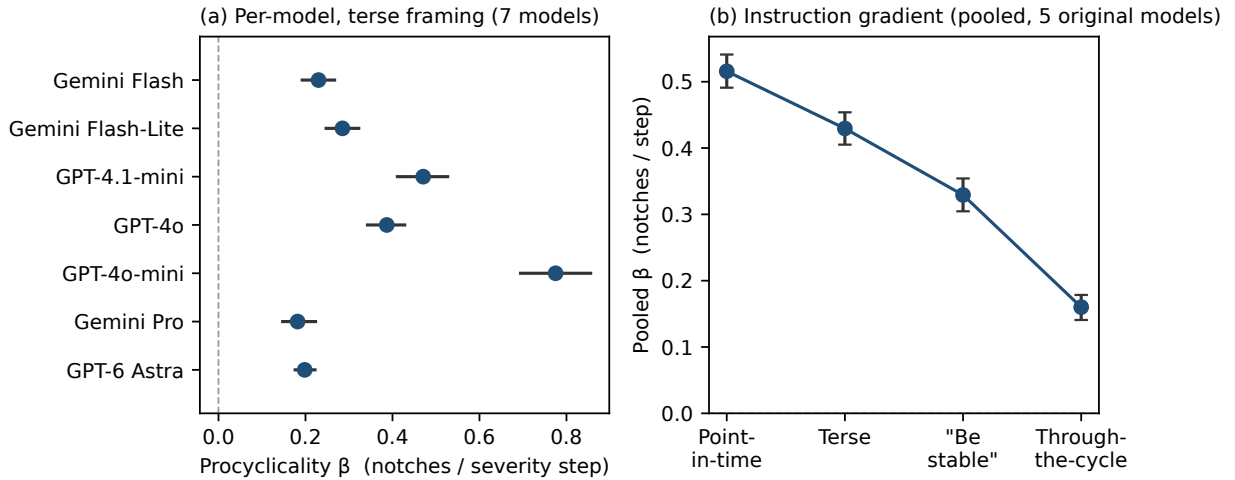
Section 3; they. They are not tools of manuscript preparation. All AI-assisted content was carefully reviewed, edited, and verified by the authors, who take full responsibility for the content of the published article.

## Declaration of competing interest

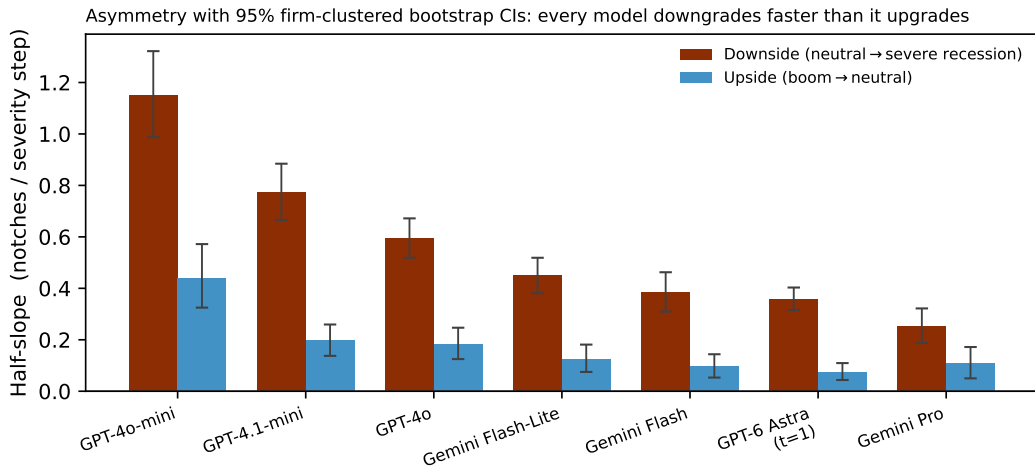
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.



**Figure 1:** (a) Per-model procyclicality  $\beta$  under the terse framing, all seven models, with 95% firm-clustered bootstrap CIs; the. The dashed line marks the through-the-cycle null ( $\beta = 0$ ). (b) Pooled  $\beta$  across the four prompt framings (point-in-time  $\rightarrow$  terse  $\rightarrow$  "be stable"  $\rightarrow$  through-the-cycle) with 95% CIs, pooled across the five original non-frontier models as in Table 2. The two frontier arms enter panel (a) and the per-model tables.



**Figure 2:** Asymmetry: per-model downside half-slope (neutral  $\rightarrow$  severe recession) versus upside half-slope (boom  $\rightarrow$  neutral), terse framing, with 95% firm-clustered bootstrap confidence intervals. Every model downgrades far-faster into recessions than it upgrades into booms, and every downside-to-upside ratio interval excludes 1. The ratio intervals are wide because the upside half-slope is small, so the ordering of models by asymmetry is not resolvable. (t=1) marks the one arm that could not run at temperature zero.

## Data availability

All data and code are openly available at in a citable Zenodo archive accompanies the published version, <https://doi.org/10.5281/zenodo.21864041> (mirrored on GitHub). Reproduction is offline, deterministic and free (Chincholikar and Chawla, 2026).

## A. Robustness

~~Robustness of the terse-framing coefficient: placebo-adjusted (net), log-probability expected-notch (OpenAI only), within-cell seed noise floor, and the Amato-Furline ordered-probit estimate. *Placebo*. Under the matched credit-irrelevant placebo axis, ratings shift by only 0.042 notches per step, so the net (macro minus placebo) effect is~~ The original confirmatory grid contains 32,000 assessments. Including the revision experiments, the study contains 80,590 frozen assessments, every one of which parsed. The rating scale has 21 points because the default grades D, SD and RD share the bottom notch with C. D accounts for 90 assessments and SD and RD for none. A further 101 call attempts, 0.13% of the 80,691 made in frozen captures, returned a transport or schema error at capture and are excluded from the frozen set. Table 3 reports the synthetic-firm generation specification.

*Placebo and alternative estimators.* The pooled macro slope is 0.4295 against 0.0425 for the matched credit-irrelevant placebo, yielding a placebo-adjusted effect of 0.387 notches per severity step (95% CI [0.36, 0.41]). ~~Infrastructure. Adding served-model (system\_fingerprint) fixed effects shifts the terse coefficient by  $< 0.003$  for every OpenAI model; a within-cell seed noise floor confirms the signal exceeds residual non-determinism.~~ [0.363, 0.412]. The interval is obtained by bootstrapping the slope difference on the same resampled firms. A joint specification containing both axes returns the same coefficient and the same interval. Served-model fixed effects shift OpenAI coefficients by less than 0.003. Table 4 reports the expected-notch, replication-noise and ordered-probit estimates.

~~ParsingReal fundamentals and issuer recall. Every model parses at 1.000, with the most cyclical model (Using FY2024 financial data from 80 SEC filers, with names, tickers, CIKs and dates removed before capture, the positive slope replicates in all four models tested: GPT-4o-mini )among the cleanest, so the effect is not a censoring or parsing artefact. Log-probabilities. For the OpenAI models, a continuous expected-notch slope from the rating-token log-probabilities reproduces the sampled estimate (e.g., GPT-4o-mini 0.775 vs 0.765).~~ *Ordered-probit.* 0.814 [0.729, 0.896], GPT-4o 0.407 [0.362, 0.451], GPT-6 Astra 0.151 [0.126, 0.179] and Gemini Flash 0.094 [0.069, 0.121]. All intervals exclude zero. Of the 45 line-by-tier comparisons between the Table 3 ranges and the real-firm interquartile ranges, 41 overlap. The four exceptions are top-tier revenue and net debt to EBITDA, and distressed-firm current ratio and working capital to assets. Because synthetic lines are drawn independently, this supports marginal credit-ratio realism, not firm scale or the joint covariance structure of real filings. GPT-6 Astra identifies 7/40 de-identified real firms and GPT-4o 0/40, versus 0/40 synthetic controls. The slope difference between identified and tested-but-unidentified firms is  $-0.003$  [ $-0.074, 0.063$ ], with the identified subsample too small to establish equivalence.

*Replication and model drift.* The primary condition has five replicates per cell for gpt-4o-mini, gpt-4.1-mini, gpt-4o and gemini-flash-latest, and two for the remaining models. Three OpenAI arms use dated snapshots (gpt-4o-mini-2024-07-18, gpt-4.1-mini-2025-04-14, gpt-4o-2024-11-20) and log system\_fingerprint, making the served backend identifiable per row but not reproducible later. GPT-6 Astra and the three Gemini -latest endpoints use unresolved aliases. Cross-epoch coefficient drift remains within sampling uncertainty.

*Formal contrasts and sensitivity.* Paired firm-clustered bootstraps show that 18 of 21 model pairs differ significantly (16 after a Holm correction). The Amato–Furfine null uses the firm mean notch as a Mundlak control and agrees in sign throughout, of comparable magnitude, though the two estimators do not perfectly preserve the cross-model rank order (Table 4). *Capital mapping.* The capital translation (§4.4) maps each notch to a stylised one-year default probability (AAA  $\approx 0.01\%$ , BBB  $\approx 0.25\%$ , BB  $\approx 1.1\%$ , B  $\approx 5\%$ , CCC  $\approx 20\%$ ; monotone across the 21-notch scale, calibrated to published agency averages) and applies the Basel IRB corporate formula (loss given default “be stable” versus through-the-cycle contrast is significant in all seven models, ranging from 0.047 [0.017, 0.080] to 0.254 [0.206, 0.304]. PD-retention intervals lie below the 0.75 threshold for every model, ranging from 0.587 [0.529, 0.642] to  $-0.106$  [ $-0.166$ ,  $-0.059$ ]. Under the “be stable” instruction PD retention instead ranges from 0.42 to 0.92.

The baseline capital mapping uses a stylised 21-notch PD ladder, anchored at approximately AAA 0.01%, BBB 0.25%, BB 1.1%, B 5% and CCC 20%, with LGD = 0.45 and maturity = 2.5 years. The ladder reproduces the ordering and broad magnitudes of the one-year corporate default rates reported by S&P Global Ratings (2025) rather than copying that study’s values. The Basel 0.05% PD input floor of Basel Committee on Banking Supervision (2020b, CRE32.4) is then applied, which binds on the four notches at AA– and above. Reported capital figures are on the floored ladder. A 259-cell sensitivity grid varies four PD mappings (baseline, PD $\times 0.5$ , PD $\times 2$  and a one-notch downward shift), three LGDs (0.30, 0.45, maturity 0.60), three maturities (1.0, 2.5, the corporate asset correlation function; no SME or firm-size adjustment) ; the reported change in capital is robust to the exact mapping. 5.0 years) and seven models, plus one verbalised-PD mapping per model. The capital swing remains positive in all 259 cells. Its magnitude varies by 2.7 to 4.7 times across the rating-based specifications and by 3.1 to 7.6 times including verbalised PD. The  $\pm 0.05$ -notch TTC rating band and 0.75 PD-retention threshold are reporting conventions rather than supervisory standards.

**Table 3**

Synthetic-firm generation specification. Each firm draws every line independently and uniformly from the interval shown for its credit-quality tier; firms are stratified across the five tiers and eight sectors so the fundamentals-implied ratings span the scale. Ranges were chosen so a tier's implied rating lands in its band, not fitted to a reference population. Of the 45 line-by-tier cells, 41 have a draw interval intersecting the interquartile range of the real firms in that tier (Appendix A). Values are frozen and SHA-256'd before capture.

| Line                       | High IG         | Low IG        | Crossover     | High yield    | Distressed     |
|----------------------------|-----------------|---------------|---------------|---------------|----------------|
| Revenue (USD m)            | 40000 to 180000 | 8000 to 60000 | 2000 to 20000 | 500 to 8000   | 100 to 3000    |
| Revenue growth (%)         | 2 to 9          | 1 to 7        | −2 to 6       | −5 to 6       | −18 to 2       |
| EBITDA margin (%)          | 22 to 38        | 15 to 27      | 11 to 20      | 7 to 16       | 0 to 9         |
| Net debt / EBITDA (×)      | 0.2 to 1.3      | 1.3 to 2.8    | 2.8 to 4.2    | 4.2 to 6.5    | 6.5 to 12.0    |
| EBITDA / interest (×)      | 14.0 to 40.0    | 6.0 to 13.0   | 3.0 to 6.0    | 1.6 to 3.2    | 0.4 to 1.6     |
| FCF margin (%)             | 9 to 18         | 4 to 10       | 0 to 6        | −4 to 3       | −14 to −1      |
| Current ratio              | 1.4 to 2.4      | 1.1 to 1.8    | 0.9 to 1.4    | 0.7 to 1.2    | 0.4 to 0.9     |
| Retained earnings / assets | 0.35 to 0.60    | 0.18 to 0.40  | 0.05 to 0.22  | −0.05 to 0.10 | −0.40 to −0.02 |
| Working capital / assets   | 0.15 to 0.32    | 0.06 to 0.20  | −0.02 to 0.12 | −0.08 to 0.06 | −0.25 to −0.02 |

**Table 4**

Robustness of the terse-framing coefficient. Net = placebo-adjusted coefficient. Exp-n. = expected-notch slope from rating-token log-probabilities. Noise SD = mean within-cell notch standard deviation across replicate seeds. Probit = Amato–Furfin ordered-probit estimate, converted from the latent scale to notches per step. Expected-notch estimates are unavailable for Gemini and for GPT-6 Astra because neither returns rating-token log-probabilities. Gemini Flash-Lite's 0.000 Noise SD is exact.

| Model             | $\beta$ | Net   | Exp-n. terse | Exp-n. TTC | Noise SD | Probit |
|-------------------|---------|-------|--------------|------------|----------|--------|
| Gemini Pro        | 0.182   | 0.174 |              |            | 0.080    | 0.138  |
| GPT-6 Astra       | 0.198   | 0.186 |              |            | 0.067    | 0.177  |
| Gemini Flash      | 0.230   | 0.190 |              |            | 0.020    | 0.272  |
| Gemini Flash-Lite | 0.285   | 0.268 |              |            | 0.000    | 0.235  |
| GPT-4o            | 0.387   | 0.367 | 0.378        | 0.182      | 0.025    | 0.403  |
| GPT-4.1-mini      | 0.471   | 0.464 | 0.420        | 0.081      | 0.086    | 0.307  |
| GPT-4o-mini       | 0.775   | 0.646 | 0.765        | 0.418      | 0.072    | 0.726  |

## B. Prompts

The prompt framings and the macro and placebo blocks below are extracted programmatically from the capture code (`agent.py`) and reproduce the executed instrument verbatim.

### B.1. Prompt framings

The four instruction framings (system prompt), verbatim as executed:

You are a credit rating analyst. Assign a long-term issuer credit rating to the company described, based on its fundamentals and the stated operating environment. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words>"}

You are a credit rating analyst applying a THROUGH-THE-CYCLE methodology. Assign a long-term issuer credit rating that reflects the company's creditworthiness over a full economic cycle, deliberately LOOKING THROUGH transitory macroeconomic conditions rather than the current point in the cycle. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words>"}

## Does AI rate through the cycle?

You are a credit rating analyst applying a POINT-IN-TIME methodology. Assign a long-term issuer credit rating that reflects the company's creditworthiness given CURRENT economic and financial conditions as described. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words">}.</p>
</div>
<div data-bbox="67 177 923 222" data-label="Text">
<p>You are a credit rating analyst. Assign a long-term issuer credit rating that is STABLE and does not change much in response to short-term conditions. Respond with ONLY a JSON object and nothing else, in exactly this form: {"rating": "<one S&P long-term issuer rating from AAA down to D, using +/- modifiers, e.g. AA-, BBB+, B>", "pd\_1yr\_bps": <your estimated 1-year probability of default in BASIS POINTS as an integer, where 100 bps = 1%>, "rationale": "<=30 words">}.</p>
</div>
<div data-bbox="67 260 386 276" data-label="Section-Header">
<h3>B.2. Macro-environment blocks</h3>
</div>
<div data-bbox="95 276 894 292" data-label="Text">
<p>Prepended to the byte-identical fundamentals; only these five blocks differ across the severity axis:</p>
</div>
<div data-bbox="67 327 472 340" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="81 338 424 394" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +4.5%</li>
<li>- Unemployment rate: 3.4%</li>
<li>- Investment-grade corporate credit spread: 95 bps</li>
<li>- Trailing 12-month corporate default rate: 0.6%</li>
<li>- Bank lending standards: easing broadly</li>
</ul>
</div>
<div data-bbox="67 437 472 450" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="81 448 430 504" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +2.6%</li>
<li>- Unemployment rate: 4.2%</li>
<li>- Investment-grade corporate credit spread: 135 bps</li>
<li>- Trailing 12-month corporate default rate: 1.4%</li>
<li>- Bank lending standards: modestly easing</li>
</ul>
</div>
<div data-bbox="67 548 472 560" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="81 559 430 613" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): +1.8%</li>
<li>- Unemployment rate: 5.0%</li>
<li>- Investment-grade corporate credit spread: 175 bps</li>
<li>- Trailing 12-month corporate default rate: 2.3%</li>
<li>- Bank lending standards: unchanged</li>
</ul>
</div>
<div data-bbox="67 657 472 669" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="81 668 430 723" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): -0.8%</li>
<li>- Unemployment rate: 6.4%</li>
<li>- Investment-grade corporate credit spread: 320 bps</li>
<li>- Trailing 12-month corporate default rate: 4.1%</li>
<li>- Bank lending standards: tightening</li>
</ul>
</div>
<div data-bbox="67 767 472 779" data-label="Text">
<p>MACRO ENVIRONMENT (current conditions, this fiscal period):</p>
</div>
<div data-bbox="81 777 430 832" data-label="List-Group">
<ul>
<li>- Real GDP growth (annualized): -3.6%</li>
<li>- Unemployment rate: 8.9%</li>
<li>- Investment-grade corporate credit spread: 620 bps</li>
<li>- Trailing 12-month corporate default rate: 7.5%</li>
<li>- Bank lending standards: sharply tightening</li>
</ul>
</div>
<div data-bbox="67 869 529 887" data-label="Section-Header">
<h3>B.3. Placebo blocks (credit-irrelevant control)</h3>
</div>
<div data-bbox="95 887 653 903" data-label="Text">
<p>Matched in structure and number density, on a credit-irrelevant axis:</p>
</div>
<div data-bbox="67 943 523 959" data-label="Page-Footer">
<p>S. Chincholikar and R. Chawla: Preprint submitted to Elsevier</p>
</div>
<div data-bbox="822 943 939 959" data-label="Page-Footer">
<p>Page 12 of 14</p>
</div>

## Does AI rate through the cycle?

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.6 degC
- Daylight hours vs annual mean: +4.5%
- Monthly rainfall vs normal: 95 mm
- Regional air-quality index: 38
- Commuter traffic congestion: light

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: +0.2 degC
- Daylight hours vs annual mean: +2.6%
- Monthly rainfall vs normal: 135 mm
- Regional air-quality index: 52
- Commuter traffic congestion: moderate

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: 0.0 degC
- Daylight hours vs annual mean: +1.8%
- Monthly rainfall vs normal: 175 mm
- Regional air-quality index: 60
- Commuter traffic congestion: typical

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -0.4 degC
- Daylight hours vs annual mean: -0.8%
- Monthly rainfall vs normal: 320 mm
- Regional air-quality index: 88
- Commuter traffic congestion: heavy

REGIONAL CONDITIONS (current, at the company's headquarters region):

- Average temperature vs seasonal norm: -1.1 degC
- Daylight hours vs annual mean: -3.6%
- Monthly rainfall vs normal: 620 mm
- Regional air-quality index: 140
- Commuter traffic congestion: severe

## CRedit authorship contribution statement

**Samir Chincholikar:** Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Robin Chawla:** Conceptualization, Validation, Writing – review and editing, Supervision.

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